HUA TONG

email ⋄ homepage ⋄ github

RESEARCH INTERESTS

mesh generation, surface reconstruction, neural operator

EDUCATION

Carnegie Mellon University *09/2022 - 01/2027*
advisor: Prof. Yongjie Jessica Zhang
Ph.D. candidate in Mechanical Engineering
GPA: 4.00/4.00

University of Science and Technology of China *08/2018 - 08/2022*
bachelor of natural science at School of the Gifted Young
major: theoretical and applied mechanics (in the Hsue-shen Tsien mechanical science and technology talent program)
GPA: 3.86/4.30 (rank 4/129)

PUBLICATION

H. Tong, Y. J. Zhang. Planar-faced and high-Jacobian two-refinement hexahedral templates. Proceedings of the 2027 SIAM International Meshing Roundtable, under review, 2027.

H. Tong, Y. J. Zhang. MCHex: marching cubes based adaptive hexahedral mesh generation with guaranteed positive Jacobian. ACM Transactions on Graphics, 2026.

Y. Yu, J. Liu, **H. Tong**, H. Lou, Y. J. Zhang. eDDPM-Polycube: an extended diffusion-based method for hexahedral mesh and volumetric spline construction. Computer-Aided Geometric Design, 129:102605, 2026.

H. Tong, Y. J. Zhang. Element-saving hexahedral 3-refinement templates. Computer-Aided Geometric Design, 127:102550, 2026.

H. Tong, Y. J. Zhang. HexOpt: efficient and robust hexahedral mesh optimization using rectified hybrid quadratic Jacobian and geometry-aware mapping. Computer-Aided Design, 196:104073, 2026.

Y. Yu, Y. Fang, **H. Tong**, J. Liu, Y. J. Zhang. DDPM-Polycube: a denoising diffusion probabilistic model for polycube-based hexahedral mesh generation and volumetric spline construction. Engineering with Computers, 42(1):1, 2025.

Y. Yu, Y. Fang, **H. Tong**, Y. J. Zhang. DL-Polycube: deep learning enhanced polycube method for high-quality hexahedral mesh generation and volumetric spline construction. Engineering with Computers, 41(3):2069-2092, 2025.

H. Tong, Y. J. Zhang. Fast and robust hexahedral mesh optimization via augmented Lagrangian, L-BFGS, and line search. Proceedings of the 2025 SIAM International Meshing Roundtable, pp. 24-33, 2025.

H. Tong, E. Halilaj, Y. J. Zhang. HybridOctree_Hex: hybrid octree-based adaptive all-hexahedral mesh generation with Jacobian control. Journal of Computational Science, 78:102278, 2024.

H. Tong, K. Qian, E. Halilaj, Y. J. Zhang. SRL-Assisted AFM: generating planar unstructured quadrilateral meshes with supervised and reinforcement learning-assisted advancing front method. Journal of Computational Science, 72:102109, 2023.

H. Tong, J. M. Hauth, X. Huan, B. Y. Zhou, N. R. Gauger, M. C. Morelli, A. Guardone. Bayesian recurrent neural networks for monitoring rotorcraft icing from aeroacoustics time-series data. AIAA Scitech 2022 Forum, p. 2358, 2022.

PRESENTATION

H. Tong, Y. J. Zhang. Advances in grid-based all-hexahedral mesh generation: saving elements and guaranteeing positive Jacobian. Graphics and Mixed Environment Symposium Webinar 412-CAX Special Issue, online. Aug 20, 2026. Talk.
H. Tong, Y. J. Zhang. Element-saving hexahedral 3-refinement templates. International Conference on Geometric Modeling and Processing, Pilsen, Czech Republic. May 27-29, 2026. Talk.
H. Tong, Y. J. Zhang. DL-Polycube for high-quality hex mesh generation and volumetric spline construction. 18th U.S. National Congress on Computational Mechanics, Chicago, IL. Jul 20-24, 2025. Talk.
H. Tong, Y. J. Zhang. Fast and robust hexahedral mesh optimization via augmented Lagrangian, L-BFGS, and line search. International Meshing Roundtable Workshop 2025, Fort Worth, TX. Mar 3-6, 2025. Talk.
H. Tong, E. Halilaj, Y. J. Zhang. HybridOctree_Hex: hybrid octree-based adaptive all-hexahedral mesh generation with Jacobian control. International Meshing Roundtable Workshop 2024, Baltimore, MD. Mar 5-8, 2024. Poster.
H. Tong, K. Qian, E. Halilaj, Y. J. Zhang. SRL-Assisted AFM: generating planar unstructured quadrilateral meshes with supervised and reinforcement learning-assisted advancing front method. 17th U.S. National Congress on Computational Mechanics, Albuquerque, NM. Jul 23-27, 2023. Talk and poster.

AWARDS

Liang Ji-Dian fellowship *02/2026*
best paper selection (top 7), Artificial Intelligence and High-Performance Computing for Advanced Simulations Workshop at the International Conference on Computational Science 2024 *10/2025*
18th U.S. National Congress on Computational Mechanics travel award *04/2025*
International Meshing Roundtable 2025 best student paper award *03/2025*
International Meshing Roundtable 2025 travel award *03/2025*
International Meshing Roundtable 2024 best technical poster award *03/2024*
International Meshing Roundtable 2024 travel award *03/2024*
17th U.S. National Congress on Computational Mechanics travel award *04/2023*
Guo Moruo award nomination *04/2022*
honorary undergraduate scholarship *12/2021*

TEACHING EXPERIENCES

teaching assistant and guest lecturer, Carnegie Mellon University 24-658/42-640: Image-Based Computational Modeling and Analysis *09/2025*
teaching assistant and guest lecturer, Carnegie Mellon University 24-703/12-703: Numerical Methods in Engineering *01/2025*